



Course Outline

Department of Mathematics and Natural Sciences
BRAC University

MAT 110: Differential Calculus & Coordinate Geometry

A. Course General Information

Course Code:	MAT 110
Course Title:	Differential Calculus & Coordinate Geometry
Credit Hours:	3
Contact Hours:	3
Type:	Theory
Prerequisites:	None
Co-requisites:	None
Course Instructor:	...
Consultation Hours:	...

B. Course Catalog Description

Differential Calculus: Function, Domain, Range, and their Properties. Limits; continuity and differentiability; successive differentiation; Leibniz's theorem; Rolle's theorem; Mean Value Theorem; Taylor and Maclaurin theorems (finite & infinite forms); L'Hôpital's rule; partial differentiation; extrema and points of inflection; applications; optimization in several variables (directional derivatives, gradient, divergence, curl, Lagrange multipliers); linear & quadratic Taylor polynomials for functions of two variables.

Coordinate Geometry: Conic sections (parabola, ellipse, hyperbola); polar, cylindrical, spherical coordinates; conics in polar coordinates.

C. Rationale of the Course

Calculus models change and underpins prediction in science & engineering. This course builds the conceptual and computational toolkit needed for later study and applications.

D. Course Objectives

- Understand Function, Domain, Range, and their Properties, limits, continuity, differentiability, and optimization.
- Master key theorems of differential calculus and their applications.
- Compute and interpret first/higher derivatives of single- and multivariable functions.
- Develop working knowledge of 2D coordinate geometry and 3D coordinates.

E. Course Plan (Weekly Schedule)

Week	Topics	Teaching–Learning Strategy	Assessment Strategy
1	Function, Domain, Range, and their Properties. Introduction to limits; indeterminate forms via L'Hôpital; horizontal/vertical asymptotes; Squeeze Theorem.	Lecture, discussion, consultation.	Assignment, quiz, examinations
2	Continuity and differentiability.	Lecture, discussion, consultation.	Assignment, quiz, examinations.
3	Techniques of differentiation; successive differentiation; Leibniz's Theorem.	Lecture, discussion, consultation.	Assignment, quiz, examinations.
4	Extrema (max/min), Rolle's and Mean Value Theorems.	Lecture, discussion, consultation.	Assignment, quiz, examinations.
5	Taylor and Maclaurin polynomials (single variable).	Lecture, discussion, consultation.	Assignment, quiz, examinations.
Midterm Examination			
6	Partial derivatives; chain rule for partial derivatives.	Lecture, discussion, consultation.	Assignment, quiz, examinations.
7	Extrema of multivariable functions; Taylor expansion (multivariable).	Lecture, discussion, consultation.	Assignment, quiz, examinations.
8	Vector calculus: fields; gradient, divergence, curl; directional derivatives.	Lecture, discussion, consultation.	Assignment, quiz, examinations.
9	Constrained optimization: Lagrange multipliers.	Lecture, discussion, consultation.	Assignment, quiz, examinations.
10	Conic sections: parabola, ellipse, hyperbola.	Lecture, discussion, consultation.	Assignment, quiz, examinations.

Week	Topics	Teaching–Learning Strategy	Assessment Strategy
11	Polar, cylindrical, spherical coordinates; conics in polar form.	Lecture, discussion, consultation.	Assignment, quiz, examinations.
Final Examination			

G. Learning Materials

Textbooks / References:

- H. Anton, *Calculus: Early Transcendentals*, 10th ed.
- M. R. Spiegel, *Schaum's Outline: Vector Analysis* (and Intro to Tensor Analysis).
- J. Stewart, D. Clegg, S. Watson, *Calculus: Early Transcendentals*, 9th ed.

H. Assessment and Evaluation

Component	Attendance	Assignment	Quiz	Midterm	Final
Weight (%)	5	15	20	20	40

Attendance Policy:

Attendance (%)	Marks
90 and above	5
85–89	4
80–84	3
75–79	2
70–74	1
Less than 70%: Unable to sit for Final Exam	

I. Grading Policy

% of Marks	Letter Grade	Grade Points
97–100	A+	4.00
90–<97	A	4.00
85–<90	A-	3.70
80–<85	B+	3.30
75–<80	B	3.00
70–<75	B-	2.70
65–<70	C+	2.30
60–<65	C	2.00
57–<60	C-	1.70
55–<57	D+	1.30
52–<55	D	1.00
50–<52	D-	0.70
<50	F	0.00

J. Administrative Information & Course Requirements

- For a make-up midterm, submit an application through the Chair/Dean within the appropriate time window; standard procedures apply.
- Students missing 30% or more classes are not allowed to sit for the final exam.

K. Academic Integrity

Students must follow BRAC University's Code of Academic Integrity and plagiarism policy. Collaboration may involve discussion and “consulting” help, but submitted work must be one's own. Any copying (giver/receiver) results in zero on the affected work and may escalate to course failure and disciplinary action. During examinations, any collaboration is strictly prohibited.